

CHEMISTRY MODEL THEORY EXAM

Group: I and II

Max. Marks: 75

Time: 3 Hrs

PART A - (10 x 2 = 20)

Answer ALL questions.

1. If 50 ml of sample of hard water consumed 15 ml of 0.01 M EDTA, what is the hardness of the water?
2. Define Caustic embrittlement and how it can be avoided?
3. What is meant by vulcanization of rubber? What are the advantages of vulcanized rubber?
4. Write short note on fibre reinforced plastics.
5. Calculate EMF of the following cell: $\text{Zn(s)} / \text{Zn}^{2+} (0.2 \text{ M}) // \text{Ag}^+ (0.002 \text{ M}) / \text{Ag(s)}$. The standard emf of the cell is 1.54 V.
6. Write short note on Nickel-Cadmium battery.
7. Why is zinc coating on iron called as sacrificial anode?
8. Differentiate between chemical and electrochemical corrosion
9. What are eutectic point and congruent melting point?
10. In the dissociation of NH_4Cl in vacuum, the following equilibrium occurs,
 $\text{NH}_4\text{Cl(s)} \longrightarrow \text{NH}_3\text{(g)} + \text{HCl(g)}$ How many numbers of phases, components, and degree of freedom are available in this equilibrium?

PART B - (11 x 5 = 55)

UNIT I

11. (a) Explain the estimation of hardness using EDTA method. (6)
(b) How the water is desalinated by electro dialysis process? (5)

Or

12. (a) Describe softening process of water by Ion exchange method. (6)
(b) List out the boiler problems aroused in boilers. Explain any two in detail.. (5)

UNIT II

13. (a) Explain the free radical mechanism of polymerization. (5)
(b) Write note on injection and transfer moulding.

Or

14. (a) Write note on conducting polymers. (5)
(b) Give the preparation, properties and uses of butyl rubber and Bakelite (6)

UNIT III

15. (a) Discuss construction and working of Lead-acid battery. (6)
(b) What is the concentration of Ni^{2+} in the cell at 25°C , if the emf is 0.601 V.
 $\text{Ni(s)} / \text{Ni}^{2+} // \text{Cu}^{2+} (0.75 \text{ M}) / \text{Cu(s)}$. Given $E^\circ \text{Ni/Ni}^{2+} = -0.25 \text{ V}$ and $E^\circ \text{Cu}^{2+}/\text{Cu} = 0.34 \text{ V}$. (6)

Or

16. (a) What is reference electrode? Explain the working principle of calomel electrode. (6)
(b) Derive Nernst equation for Single Electrode Potential (5)

UNIT IV

17. (a) Explain mechanism of electrochemical corrosion. (6)
(b) Write a brief note on cathodic protection method. (5)

Or

18. (a) What is anodizing? Explain the process in detail. (6)
(b) Discuss Galvanic cell corrosion and Stress corrosion. (5)

UNIT V

19. (a) State phase rule and explain the terms involved in it. (5)
(b) With a neat sketch, explain Sulphur system. (6)

Or

20. (a) What is reduced phase rule? Give a brief note on thermal analysis. (5)
(b) With a neat sketch explain Zn - Mg alloy system. (6)